

# Utkarsh Verma

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## PROFESSIONAL SUMMARY

AI/ML Engineer with 1.5+ years of production Generative AI experience inside enterprise products — LLM contract intelligence, multimodal invoice automation, and a benchmarked multilingual ASR→NLP pipeline. Strong across RAG, LangGraph/LangChain, vector search, evaluation (RAGAS), and FastAPI/AWS. Also creating Xcloom, an AI social-content automation platform.

## TECHNICAL SKILLS

- Generative AI / LLMs:** RAG, LangChain, LangGraph, Hugging Face Transformers, OpenAI / GPT APIs, Prompt Engineering, LoRA Fine-tuning, Agentic Workflows, Vector DBs (FAISS, Chroma, pgvector), Model Context Protocol (MCP), RAGAS Evaluation, Semantic + Lexical Retrieval, Reranking, OCR, PDF Parsing, Entity Extraction, Summarization, Speech-to-Text (Whisper, ASR)
- Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Model Evaluation & Optimization, Feature Engineering, PyTorch, Transformers
- MLOps & Deployment:** AWS (EC2, S3), Docker, GitHub Actions, CI/CD for ML, Model Monitoring, Cloudflare R2
- Programming & Data:** Python, SQL, FastAPI, Webhooks, Pandas, NumPy, PostgreSQL, Celery

## PROFESSIONAL EXPERIENCE

### Softude Infotech Pvt. Ltd.

July 2025 – Present

Associate Software Engineer — AI/ML

Indore, India

#### Contract Intelligence — Netcore CMT

- Built a production RAG pipeline (LangGraph + GPT-4o + PostgreSQL/pgvector) auto-extracting 25+ structured contract fields (SLA, payment terms, governing law, indemnity, auto-renewal) from PDF/DOCX, cutting contract onboarding ~90% from 2–3 hrs of manual legal review to sub-minute AI processing.
- Designed hybrid semantic + lexical retrieval with LLM query rewriting, a rerank filter (top-30 → top-15 via gpt-4o-mini), and a confidence gate to block hallucinated answers — evaluated with RAGAS at >80% faithfulness and answer relevancy on held-out queries.
- Cut per-token inference cost ~16× on extraction and reranking by routing them to gpt-4o-mini and reserving gpt-4o for final synthesis, with per-operation token/cost/latency observability (AIUsageLog).

#### Invoice Automation — Netcore P2P

- Developed a multi-modal invoice intelligence system (GPT-4o Vision, text-first → PDF-rasterization fallback) extracting structured fields (vendor, buyer, GST, total, currency) across 6 regional formats — ~88% field-level accuracy, ~60% less manual data entry.
- Engineered a hybrid rule + AI validation pipeline (region detection across IN/US/UK/SG/MY/NG + gpt-4o-mini PO reconciliation, 1% mismatch tolerance) — ~70% straight-through processing; added an extraction cache cutting redundant token spend ~40% on bulk reprocessing.

#### MIS & Billing — Netcore Pulse

- Sole owner of a multi-channel MIS platform unifying SMS, Email, WhatsApp, and RCS revenue/cost/margin/projection metrics across 12,000+ daily records for 200+ users — consolidating 4 fragmented reporting streams into one schema and cutting month-end reporting prep ~65%.
- Automated SAP billing reconciliation with advance-billing lifecycle management, reducing finance reconciliation effort ~70% and giving real-time billing visibility.

#### Multilingual Call Intelligence — VECV / Volvo Eicher

- Built an end-to-end speech-to-NLP pipeline (Whisper ASR → translation → LLM intent & entity extraction) on FastAPI across 6–7 Indian languages and load-tested for 500K+ calls/month capacity.
- Measured 83% transcription accuracy on noisy telephony audio during benchmarking, structuring call context for downstream analytics (production rollout pending)

### RPS Business Solution Pvt. Ltd.

Feb 2025 – June 2025

Data Scientist Intern

Indore, India

- Spearheaded an end-to-end RAG application (LangChain + Hugging Face + FAISS) over 150+ unstructured business documents, enabling natural-language retrieval across domain-specific data.
- Fine-tuned a Llama-2-7B medical-domain assistant on the company's proprietary dataset using LoRA, lifting answer accuracy from 40% to 55% on held-out medical queries vs the base model.

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| <b>Freelance Engagements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>Sep 2024 – June 2025</i> |
| <i>Delivered AI solutions for 5+ clients across India &amp; the US (INR 2.7L+ revenue)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |
| <ul style="list-style-type: none"> <li>• <b>Neural Hire (AI Recruitment):</b> Solo- developed a recruitment platform — LLM PDF-to-job-post generation (raw docs → publish-ready LinkedIn descriptions + AI visuals) producing a finished post in seconds vs ~30 min of manual writing, plus candidate shortlisting and a WhatsApp bot (Graph API) for two-way outreach, automating the post-to-outreach flow end-to-end.</li> <li>• <b>Fring (AI Social Matching):</b> Proposed a hybrid matching engine combining LLM relevance scoring with embedding similarity into a tunable weighted score, reducing low-quality match suggestions by 67% vs the similarity-only baseline.</li> <li>• <b>Call-Insights RAG (US sales calls):</b> Shipped a RAG system (LangChain, Chroma, Whisper, Streamlit) over 200+ diarized US sales calls, enabling natural-language querying of call insights.</li> </ul> |                             |

KEY PROJECTS

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| <b>Xcloom — AI-Powered Social Media Automation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>2026 – Present   Personal Project</i> |
| <i>End-to-end product: AI image generation + Instagram scheduling automation</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                          |
| <ul style="list-style-type: none"> <li>• Architected an 8-stage LangGraph generation pipeline (context loading → extraction → vision analysis → strategy → prompt → image → caption → assembly) producing publish-ready social posts end-to-end from a single prompt.</li> <li>• Created GPT-4o caption chains (LangChain) with purpose-conditioned prompts (creator / brand / educator) and structured JSON parsing; integrated DALL·E 3 for image synthesis and GPT-4o Vision for reference-image style analysis.</li> <li>• Implemented an MCP tool server exposing 5 custom AI tools (content extraction, prompt generation, scheduling, metadata, external HTTP); orchestrated async job (Celery + FastAPI) with status tracking, retries, and fallbacks.</li> <li>• Engineered resilient Instagram Graph API publishing (container-status polling, media recovery); secured via Fernet-encrypted OAuth tokens, JWT auth, and async SQLAlchemy on PostgreSQL, serving a Flutter client.</li> </ul> |                                          |

EDUCATION

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|----------------------------------------|-------------------------------|
| <b>Medi-caps University</b>            | <i>Aug 2022 - June 2024</i>   |
| <i>Master of Computer Applications</i> | <i>Indore, Madhya Pradesh</i> |
| <b>RPL Maheshwari College (DAVV)</b>   | <i>Aug 2019 - May 2022</i>    |
| <i>Bachelor of Commerce</i>            | <i>Indore, Madhya Pradesh</i> |

CERTIFICATION & CONTINUOUS LEARNING

Master Program in Artificial Intelligence & Machine Learning – Bignalytics

Machine Learning: AI, Python — Udemy